

Q1: BitTorrent Case Study (Short Answers)

- How BitTorrent Works:**
BitTorrent splits large files into smaller pieces and distributes them among peers. Each peer downloads and uploads pieces simultaneously, unlike traditional file-sharing, which relies on a single server.
 - Seeders and Leechers:**
 - **Seeders:** Users who upload file pieces to the network.
 - **Leechers:** Users who download file pieces but may also upload simultaneously.
They ensure efficient file sharing by distributing the load.
 - Torrent File:**
A small file containing metadata (file name, size, and tracker information) that helps connect peers to share the larger file.
 - Advantages & Disadvantages:**
 - **Advantages:** Faster downloads, decentralized, efficient for large files.
 - **Disadvantages:** Requires active peers, potential copyright issues, and slower if few seeders.
 - Data Integrity & Reliability:**
File parts include checksums to verify correctness. If a part is corrupted, BitTorrent re-downloads it from another peer.
 - Copyright Measures:**
BitTorrent itself doesn't regulate content, but legal distributors can use encryption, authentication, or copyright notices to protect their work.
 - Challenges & Solutions:**
Challenges include ISP throttling, slow speeds, and piracy. Developers have added encryption, better trackers, and decentralized methods like DHT.
 - Protocol Evolution:**
Enhanced peer discovery (DHT), magnet links replacing torrent files, and encryption have improved performance and security.
 - Decentralized Tracking:**
Uses Distributed Hash Tables (DHT) instead of central trackers, improving scalability and reducing single points of failure.
 - Alternative Uses:**
Efficient software distribution, blockchain applications, and decentralized cloud storage platforms.
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Q2: Netflix Case Study (Short Answers)

- Impact on TV & Entertainment:**
Shifted focus from scheduled programming to on-demand streaming, reducing reliance on traditional cable TV.
- Key Success Factors:**
Original content, global availability, user-friendly interface, and personalized recommendations.
- Algorithms & Data Analytics:**
Analyze viewing patterns to recommend shows, boosting user engagement and retention.
- Original Content Impact:**
Revolutionized global content production with series like *Stranger Things*, fostering diverse storytelling.
- Regional Restrictions:**
Uses licensing agreements and VPN detection to manage content availability by region.
- Binge-Watching Trend:**
Encouraged continuous viewing by releasing entire seasons at once, reshaping viewer habits.
- Role of Technology:**
Adaptive streaming adjusts quality based on internet speed, ensuring smooth playback on various devices.
- Data Privacy:**
Employs encryption and anonymization to secure user data while using it for personalized recommendations.
- Competition Effects:**
Faces challenges from Disney+, Amazon Prime, etc., but innovates with better content, pricing, and features.
- Content Discovery & Retention:**
Uses AI for recommendations and introduces new content regularly to keep users engaged.

